

Manage those fans in a better way and easily cut down power consumption

Power and automation are two areas that stand to see significant gains from modern sensor, processing and communication technologies. Vivek Sharma, regional vice president, Greater China and South Asia region - India operations, director - India design centres, STMicroelectronics, speaks with Dilin Anand of EFY



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Q. What aspect of power the engineers should know?

A. Smartpower is a very important aspect. Power generation has both conventional and non-conventional aspects. Many electronics are going into non-conventional areas. Solar, for example, is a solution that ensures there is no reduction in power consumption even if a part of the panel comes under shade. When you look at power inverters, it is a need specific to our country and not something seen on the international stage.

Power-line communications is another technology that works in the international arena but not in India. When it comes to transmission of power in India, we end up using power even when we are transmitting.

Pilferage is one aspect; non-efficient way of transmission is another that lets you optimise these areas. Power-line communications and smartmetering can help identify where the leakages are happening.

Q. What are some ways to reduce power consumption?

A. Motor control is an important area here. Motors worldwide consume more than 55 per cent of power. If we can just have a better, more efficient way of controlling motors, like in a fan, we could save a lot of power.

We have millions of fans being used every year. If we had those fans with more electronics, call it electronic fans, we could manage those fans in a better way and easily cut down power consumption by 40 per cent. This can be done by utilising the latest sensor, microcontroller and communication technologies.

Q. What makes sensor technology so important?

A. A sensor like an accelerometer is not something new. But, it has been miniaturised and the volume of production has made it so cost-effective that you can put it almost anywhere, including toys, wearables and even vehicles.

Q. How has electronics helped improve vehicular safety?

A. Proactive safety is where we try to prevent the event from happening, and electronics plays an important role here. Radar systems or driver-assist systems are examples where drivers are warned according to the data collected by electronics systems if something somewhere

has a chance of going wrong. What happens is that, the system alerts the driver, hence preventing the accident.

We also have anti-skid features where the system detects a skid and implements measures to stop the vehicle from skidding, thus preventing accidents.

Q. Automotive infotainment seems to be a very exciting segment too; what is happening there?

A. Automotive infotainment has migrated from what used to be a CD player to something that provides navigation in your car, like a GPS, and connected cars.

You no longer just monitor car conditions but also transmit these online. This means, if you have given your car to a friend, you can see what is happening apart from where the car is. You can also see how the driver is driving. This is very helpful for people who are running fleets of buses or trucks. Fuel consumption could also be monitored and optimised through this. So, connected vehicles are expected to be very important.

Q. What is exciting thing about industrial Internet?

A. Industrial Internet is a beautiful thing. An industrial cycle is like a supply chain; you can let these devices communicate among themselves, thus reducing the burden on humans. It can help enable parts requirement and production management to be done by computers, thus speeding up the process and enabling humans to focus on more intellectual challenges. For instance, if a machine's health is deteriorating, you can have the machine communicate to you when its health hits a pre-set low, and it will let you know exactly when to interfere.

Q. How is automation we previously had different from what we have now?

A. Let us take an example of an air-conditioner at home. When it is running, the thermostat is communicating with the air-compressor. It checks the room temperature and when the temperature goes down a particular limit, it stops the compressor. This was previously happening within the machine, but now the Internet flattens everything and allows all machines to come on to the same platform. ●